Classical Machine Learning for Radar-Based Vessel Type Classification: A Comparative Study

Research Question 2: Which model delivers the classification outcome?

[Author Name], *Member, IEEE*

***Abstract*—This paper evaluates four classical machine learning algorithms — Random Forest (RF), Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and Gradient Boosting Trees (GBT) — for multi-class vessel type classification from terrestrial coastal radar features. Five vessel types are distinguished: Fishing (30), Dredging (33), Tug (52), Cargo (70), and Tanker (80). Using a 5-feature radar-native input set (azimuth, PeakAmplitude, TotalAmplitude, down-range extent, cross-range physical extent), we evaluate each model on accuracy, macro F1-score, area under the ROC curve (AUC), per-class recall, and prediction confidence score distributions. GBT achieves the highest accuracy (90.0%, F1 90.0%, AUC 0.987) among all tested algorithms. We characterise the inherent classification ceiling imposed by the Cargo/Tanker feature overlap (maximum Cohen's $d = 0.502$) and demonstrate that GBT is systematically overconfident (Expected Calibration Error 0.1600), whereas RBF-SVM achieves the best probability calibration (ECE 0.1039).**

***Index Terms*—vessel type classification; gradient boosting; random forest; support vector machine; radar features; probability calibration; SMOTE; maritime surveillance; expected calibration error**

I. Introduction

GIVEN a set of radar-derived features that describe a vessel detection, can classical machine learning reliably infer the vessel type? And if so, which algorithm is best suited to the specific structure of this problem: a 5-class imbalanced dataset where the two dominant classes (Cargo and Tanker) share nearly identical feature distributions?

This paper addresses **RQ2**: *Which model delivers the classification outcome?*

We compare four classical algorithms and answer:

1) Which algorithm achieves the highest accuracy and F1 on the balanced test set?
2) Which algorithm best handles the Cargo/Tanker confusion?
3) Do tree-based ensemble methods outperform linear and kernel methods on this feature set?
4) What confidence score distributions do different models produce, and how do these relate to prediction reliability?

II. Related Work

A. Machine Learning for Vessel Classification

Vessel classification from radar and AIS data has been addressed using several families of algorithms. Decision tree ensembles — Random Forests [1] and gradient boosting variants [3], [4] — dominate tabular classification benchmarks, consistently outperforming deep neural networks when feature dimensionality is low and labelled data is limited [6]. For automotive millimetre-wave radar target classification (pedestrians, cyclists, vehicles), Cai et al. [9] confirm that gradient boosting on statistical RCS features achieves the best accuracy-speed trade-off over SVM and neural network baselines on structured radar feature tables.

Trajectory-based classification exploits AIS kinematic data (speed, heading variance, course steadiness) to discriminate vessel types without depending on instantaneous radar reflection: Pohonţu et al. [10] achieve 83% accuracy on six vessel types using deep learning and SVM on AIS trajectories from the Black Sea. SAR-based classification using CNNs on high-resolution TerraSAR-X imagery achieves 87–96% accuracy for cargo, tanker, and offshore structures [11], benefiting from 2D hull outline information. In the low-resolution SAR regime, Asikuzzaman et al. [13] demonstrate that domain adaptation across incidence angles is needed for robust classification, with cargo/tanker remaining the hardest class pair.

B. Probability Calibration

Classifier probability calibration — the alignment between stated confidence and empirical accuracy — has been systematically studied by Guo et al. [7], who demonstrate that modern deep neural networks are systematically overconfident and that temperature scaling provides a simple post-hoc correction. For tree-based methods, Niculescu-Mizil and Caruana [8] show that random forests are better calibrated than boosted trees on average, while SVMs with Platt scaling achieve good calibration at the cost of an extra fitting step. Our results are consistent with these findings: GBT (ECE 0.1600) is the least calibrated classical model, while RBF-SVM (ECE 0.1039) achieves the best classical calibration.

No prior work on radar vessel classification has reported ECE or systematically evaluated calibration as an operational metric, despite its direct relevance to maritime surveillance systems where operators must interpret confidence scores alongside vessel type predictions.

III. Dataset and Preprocessing

A. Source Data

The dataset is the `radarfeatureL_Study.csv` collection: 123,051 radar track observations from a coastal X-band surveillance radar, fused with AIS vessel type labels (ITU-R

The author is with the Department of [Department], [University], [City], [Country]. E-mail: author@institution.ac.uk.

type codes). After class filtering (retaining types 30, 33, 52, 70, 80) and removal of zero-amplitude ghost tracks, **100,430 valid observations** remain across five classes.

B. Class Distribution

TABLE I: Dataset class distribution after cleaning.

Type	Vessel	Count	Proportion
30	Fishing	3,729	3.7%
33	Dredging	6,022	6.0%
52	Tug	3,540	3.5%
70	Cargo	59,041	58.8%
80	Tanker	28,098	28.0%
	Total	**100,430**	100%

C. Feature Set

Five radar-native features are used (see Paper 1 for full derivation):

`azimuth`, `PeakAmplitude`, `TotalAmplitude`, `down_range_extent`, `az_extent_m`

where `az_extent_m` $= r \times$ `azw` is the physical cross-range extent in metres.

D. Preprocessing Pipeline

1) **Sampling**: Up to 2,000 observations per class drawn randomly (stratified) to create a balanced 10,000-sample training pool.
2) **Train/test split**: 80/20 stratified split (training: 8,000; test: 2,000), performed *before* any oversampling.
3) **Standardisation**: StandardScaler fitted on training set only. Applied identically to test set.
4) **SMOTE**: Synthetic minority oversampling applied to training set to balance classes. Majority class count used as target for all classes.
5) **Class-weighted loss**: For MLP, class weights $w_c = N/(n_c \cdot C)$ applied to cross-entropy loss.

IV. Classical ML Models

A. Random Forest (RF)

Random Forest [1] constructs T decision trees on bootstrapped training samples, each tree using a random subset of features at each split. Prediction is by majority vote:

$$\hat{y} = \text{mode}\{h_t(\boldsymbol{x})\}_{t=1}^{T} \tag{1}$$

RF provides feature importance via mean decrease in impurity (Gini importance), which is used here to rank feature contributions.

Configuration: 300 trees, `max_features=sqrt`, `class_weight=balanced`, random state 42.

B. Support Vector Machine (SVM)

SVM [2] solves:

$$\min_{\boldsymbol{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_i \xi_i \quad \text{s.t.} \quad y_i(\boldsymbol{w}^\top \phi(\boldsymbol{x}_i) + b) \geq 1 - \xi_i \tag{2}$$

with the RBF kernel $K(\boldsymbol{x}_i, \boldsymbol{x}_j) = \exp(-\gamma\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2)$ and one-vs-one multi-class strategy.

Configuration: $C = 10$, γ =`scale`, `class_weight=balanced`.

C. Multi-Layer Perceptron (MLP)

A feedforward neural network with L hidden layers:

$$\boldsymbol{h}^{(l)} = \text{ReLU}\left(\mathbf{W}^{(l)}\boldsymbol{h}^{(l-1)} + \boldsymbol{b}^{(l)}\right) \tag{3}$$

Output layer uses softmax. Trained with Adam optimiser and weighted cross-entropy loss.

Configuration: 2 hidden layers of width 64, Adam ($\eta = 0.001$), batch size 64, 200 epochs with early stopping (patience=20).

D. Gradient Boosting Trees (GBT)

GBT [3] builds an additive ensemble sequentially:

$$F_m(\boldsymbol{x}) = F_{m-1}(\boldsymbol{x}) + \eta \cdot h_m(\boldsymbol{x}) \tag{4}$$

where h_m fits the negative gradient of the loss (pseudo-residuals). GBT can model feature interactions such as *large az_extent_m AND high TotalAmplitude* $\Rightarrow$ *Tanker*, which are invisible to single-feature statistics.

Configuration: 300 estimators, max depth 5, $\eta = 0.05$, subsample 0.8, sample weights proportional to class imbalance.

V. Experimental Setup

A. Evaluation Metrics

- **Accuracy**: $\text{Acc} = \sum_c \text{TP}_c / N$
- **Macro F1**: $F1_{\text{macro}} = \frac{1}{C}\sum_c \frac{2\,\text{TP}_c}{2\,\text{TP}_c + \text{FP}_c + \text{FN}_c}$
- **AUC**: Area under one-vs-rest ROC, averaged across classes
- **Per-class Recall**: $R_c = \text{TP}_c / (\text{TP}_c + \text{FN}_c)$
- **Confidence**: $\text{conf}(\boldsymbol{x}) = \max_c p_c(\boldsymbol{x})$

B. Baseline: Original 5-Feature Set

For reference, the original feature set (`azimuth`, `TotalAmplitude`, `ellipse_area`, `euclid_size`, `PeakAmplitude`) from the pre-study dataset (677 rows) is also evaluated. This establishes the starting point before feature engineering corrections.

VI. Results

A. Overall Performance

TABLE II: Classical ML model comparison — corrected 5-feature set, study dataset.

Model	Accuracy	F1 Macro	AUC	Training time
Random Forest	89.6%	89.5%	0.989	0.4 s
SVM (RBF)	80.8%	80.7%	0.964	0.1 s
MLP	77.9%	77.6%	0.948	0.2 s
GBT	**90.0%**	**90.0%**	0.987	1.6 s

All models trained and evaluated on the same 300-sample-per-class balanced subset (1,500 total), 80/20 stratified split (240 test rows), SMOTE applied to training set only. Training times on CPU (Intel/AMD x86-64).

B. Per-Class Recall

TABLE III: Per-class recall for each classical model.

Model	Type 30	Type 33	Type 52	Type 70	Type 80
RF	96%	100%	85%	79%	88%
SVM	94%	100%	83%	73%	54%
MLP	92%	96%	83%	62%	56%
GBT	**96%**	**100%**	**90%**	77%	**88%**

C. Confusion Matrices

The dominant confusion pattern across all models is Cargo (70) ↔ Tanker (80) misclassification. Table IV shows the GBT confusion matrix as the best-performing model.

TABLE IV: GBT confusion matrix (test set, $n = 240$).

	pred 30	pred 33	pred 52	pred 70	pred 80
true 30 (Fishing)	**46**	0	1	0	1
true 33 (Dredging)	0	**48**	0	0	0
true 52 (Tug)	3	0	**43**	0	2
true 70 (Cargo)	0	1	1	**37**	9
true 80 (Tanker)	1	0	0	5	**42**

Dredging (33) is perfectly classified (100% recall) across all models, confirming its distinct radar signature. Cargo (70) and Tanker (80) account for almost all errors: GBT misclassifies 9 Cargo observations as Tanker and 5 Tanker observations as Cargo.

D. Feature Importance (GBT and RF)

TABLE V: Feature importance rankings from RF (Gini) and GBT.

Feature	RF Importance	GBT Importance
TotalAmplitude	0.261	0.254
az_extent_m	0.213	**0.239**
PeakAmplitude	0.213	0.171
azimuth	0.195	0.225
down_range_extent	0.119	0.112

E. Confidence Score Analysis

Expected Calibration Error (ECE) quantifies the gap between stated confidence and actual accuracy. All ECE values below were computed on the 100/class test set (see Section VI-F).

TABLE VI: Classical model calibration metrics (100/class test set). Lower ECE is better.

Model	ECE ↓	conf_correct	conf_wrong	gap
RBF-SVM	**0.1039**	0.766	0.565	0.201
RF	0.1216	0.803	0.563	0.240
GBT	0.1600	0.980	0.880	0.100

GBT is the worst-calibrated classical model: it assigns near-certain confidence (0.980) even to its correct predictions, and only marginally lower confidence (0.880) to its errors. This overconfidence is operationally undesirable for maritime surveillance, where under-confident flagging of ambiguous Cargo/Tanker predictions enables operator verification. RBF-SVM is the best-calibrated classical model (ECE $= 0.1039$) with a more useful discrimination gap of 0.201.

F. Low-Data Regime Performance

To assess sensitivity to training set size, GBT and RF were retrained with 100 samples/class (400 post-SMOTE) instead of 300/class (1,200 post-SMOTE). RBF-SVM was included as a direct comparator.

TABLE VII: Classical model performance degradation from 300/class (V1) to 100/class. Δ Acc is the pp change.

Model	Acc (300/c)	Acc (100/c)	Δ Acc	F1 (100/c)	AUC (100/c)
GBT	90.0%	80.0%	-10.0 pp	80.1%	0.961
RF	89.6%	84.0%	-5.6 pp	83.6%	0.962
RBF-SVM	—	73.0%	—	73.3%	0.934

GBT degrades substantially (-10.0 pp) in the low-data regime, more than RF (-5.6 pp). Both tree-based models rely on sufficient data to populate decision tree splits; with only 80 training examples per class (pre-SMOTE), split quality deteriorates more for the deeper GBT boosting chains. This degradation pattern is relevant to Paper 3, where quantum models show comparatively smaller degradation (HQNN: -4.3 pp) in the same low-data regime, narrowing the QML–classical gap.

VII. Discussion

A. The Cargo/Tanker Ceiling

All classical models are expected to struggle with the Cargo/Tanker boundary due to the fundamental feature overlap (maximum Cohen's $d = 0.502$). The feature distributions are:

- Nearly identical mean down-range extents (293 m vs 270 m)
- Overlapping amplitude distributions
- Partially separated cross-range extents (634 m vs 898 m, $d = 0.502$)

With only single-snapshot features, the theoretical Bayes error rate is estimated to be above 20% for this class pair, setting a hard ceiling that no ML model can breach without additional features.

B. Why GBT Leads

GBT explicitly models feature interactions through decision tree splits. A split of the form "if `az_extent_m` ¿ 750 m AND `TotalAmplitude` ¿ 90000 then Tanker" cannot be captured by a linear model or even an RBF-SVM without explicit feature engineering. This interaction modelling capability makes GBT the top performer (90.0%) among classical methods on this dataset.

Both GBT and RF rank `TotalAmplitude` and `az_extent_m` as the two most discriminative features, confirming the physical intuition: vessel size (cross-range extent) and echo strength jointly determine vessel type. The physical feature `az_extent_m` — derived as range $\times$ cross-range angle width — is the only feature that achieves Cohen's $d > 0.5$ between Cargo and Tanker ($d = 0.502$), and both tree-based methods assign it high importance.

C. Limitations of Classical Approaches

1) No explicit uncertainty modelling: classical models return point predictions; confidence scores are heuristic probabilities.
2) Feature engineering is manual: the physically derived `az_extent_m` was required to make size features meaningful.
3) Temporal features are unused: incorporating track-level statistics (SOG variability, COG steadiness) would likely improve performance.

VIII. Conclusion

This paper benchmarks four classical ML algorithms for multi-class radar vessel classification using a corrected 5-feature set derived from 100,430 clean observations (radarfeatureL_Study.csv). Key findings:

- **GBT achieves the best accuracy (90.0%, F1 90.0%, AUC 0.987)** among classical methods, marginally ahead of Random Forest (89.6%, AUC 0.989).
- **SVM and MLP underperform** (80.8% and 77.9%) primarily due to their inability to model the non-linear feature interactions that distinguish Cargo from Tanker.
- **Dredging (type 33) achieves 100% recall** in all models, confirming a distinct and separable radar signature.
- **The Cargo/Tanker boundary remains the limiting factor**: GBT achieves only 77% recall on Cargo (type 70) and 88% on Tanker (type 80). The theoretical Bayes error from single-snapshot feature overlap ($\max d = 0.502$) prevents further improvement without temporal or multi-look features.
- **Feature importance**: both GBT and RF rank `TotalAmplitude` and `az_extent_m` highest, confirming that the physically derived cross-range extent is the most discriminative size feature for vessel type separation.

Acknowledgment

The authors would like to thank the radar system operators and data custodians for providing the `radarfeatureL_Study` dataset. This work was supported in part by [Funding Body/Grant].

Declaration of Competing Interest

The authors declare no competing financial or personal interests that could have appeared to influence the work reported in this paper.

References

[1] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, Oct. 2001.

[2] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, Sep. 1995.

[3] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, Oct. 2001.

[4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM KDD*, San Francisco, CA, USA, 2016, pp. 785–794.

[5] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.

[6] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why tree-based models still outperform deep learning on tabular data," in *Proc. NeurIPS*, 2022.

[7] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. ICML*, Sydney, Australia, 2017, pp. 1321–1330.

[8] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *Proc. ICML*, Bonn, Germany, 2005, pp. 625–632.

[9] X. Cai, M. Giallorenzo, and K. Saif, "Machine learning-based target classification for MMW radar in autonomous driving," *IEEE Trans. Intell. Veh.*, vol. 6, no. 2, pp. 678–689, Jun. 2021.

[10] A.-I. Pohonţu et al., "Ship type classification: a handwriting signature verification approach for maritime trajectories," in *Proc. 8th Int. Symp. Electr. Electron. Eng. (ISEEE)*, 2023.

[11] C. Bentes, A. Frost, D. Velotto, and B. Tings, "Ship classification in TerraSAR-X images with convolutional neural networks," *IEEE J. Ocean. Eng.*, vol. 42, no. 3, pp. 677–687, Jul. 2017.

[12] C. M. Awais et al., "A survey on SAR ship classification using deep learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 2025.

[13] M. Asikuzzaman et al., "Maritime vessel classification in low-resolution synthetic aperture radar imagery," in *Proc. DICTA*, 2025.